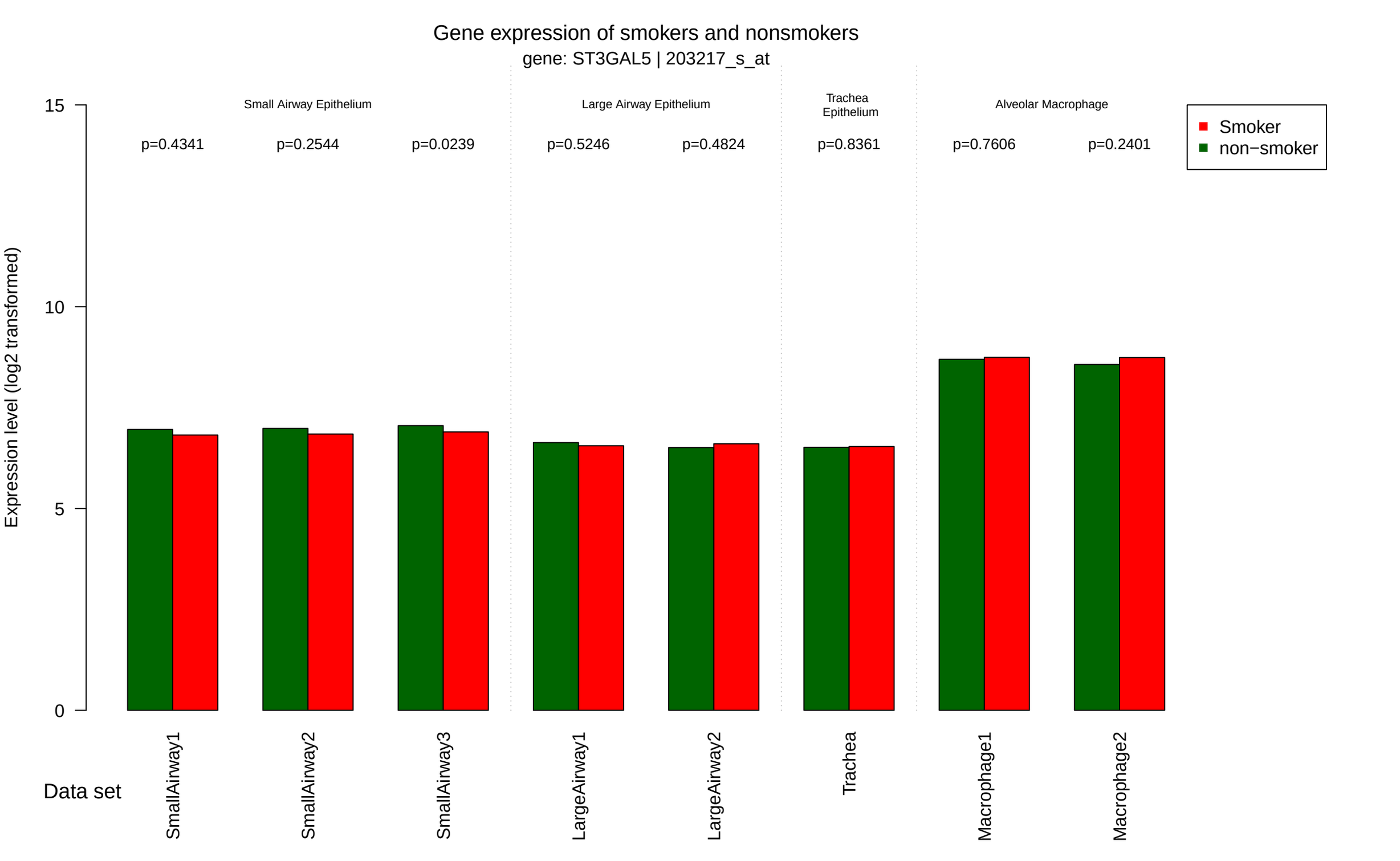
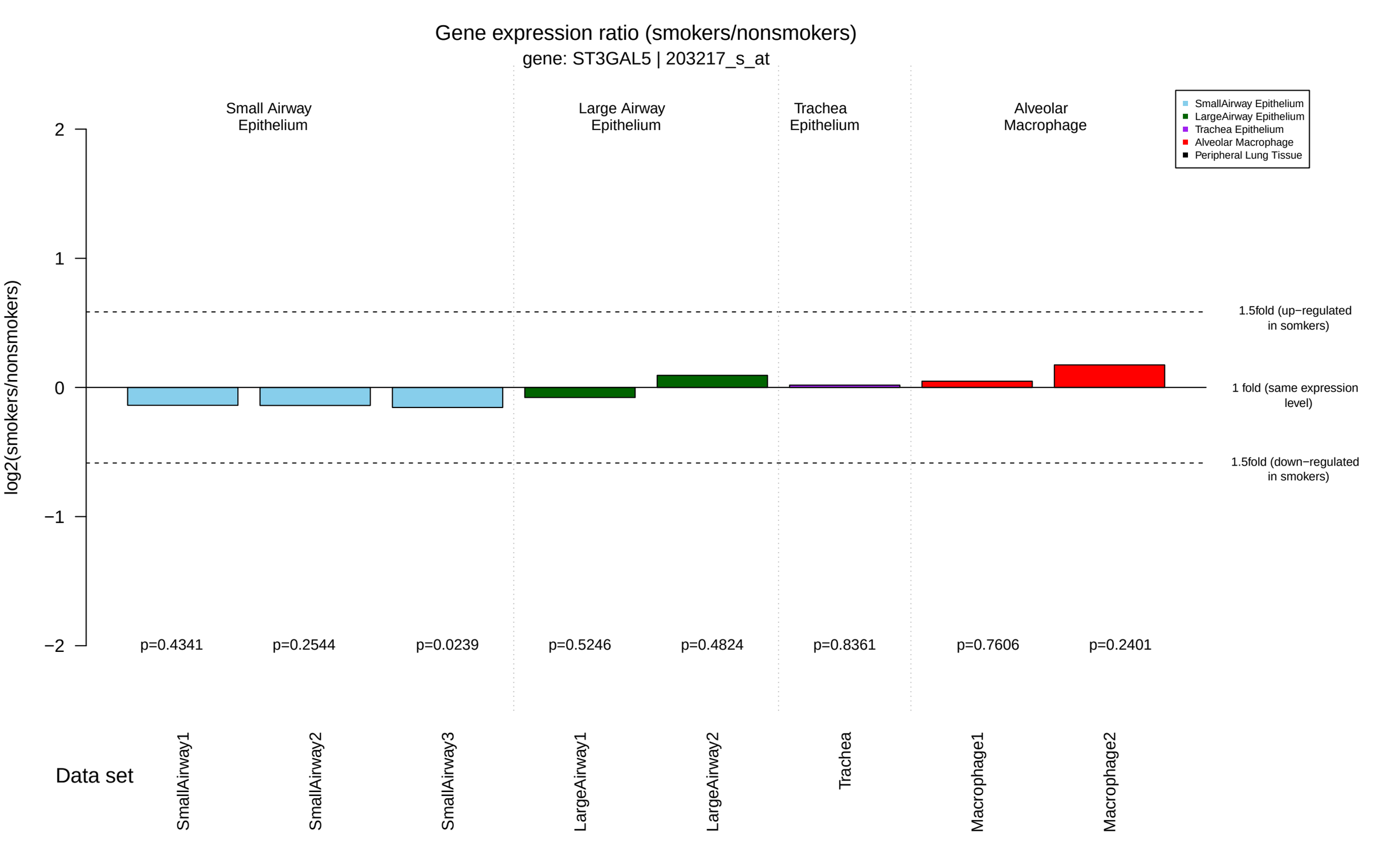
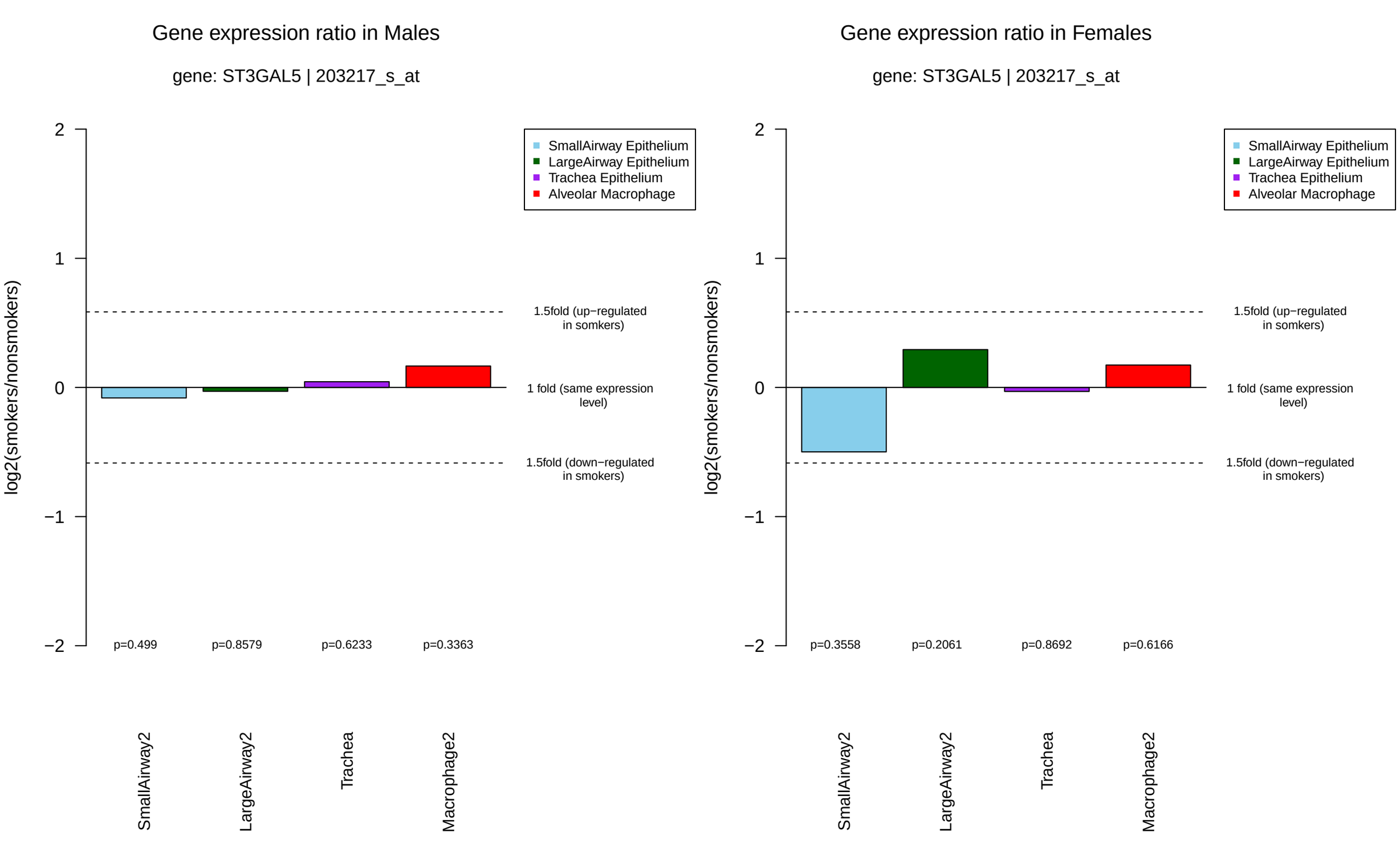
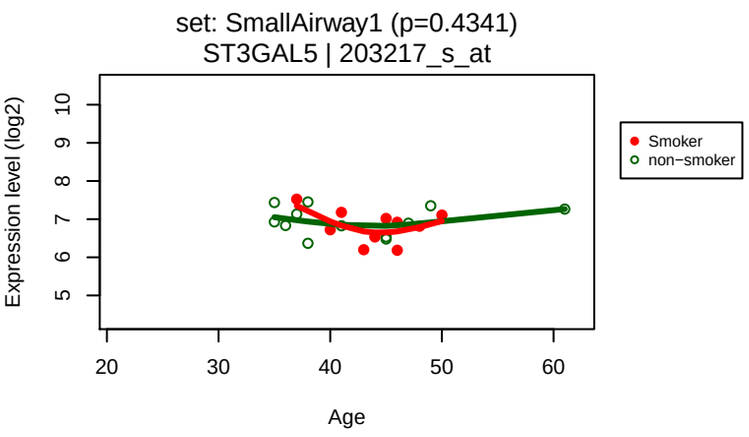
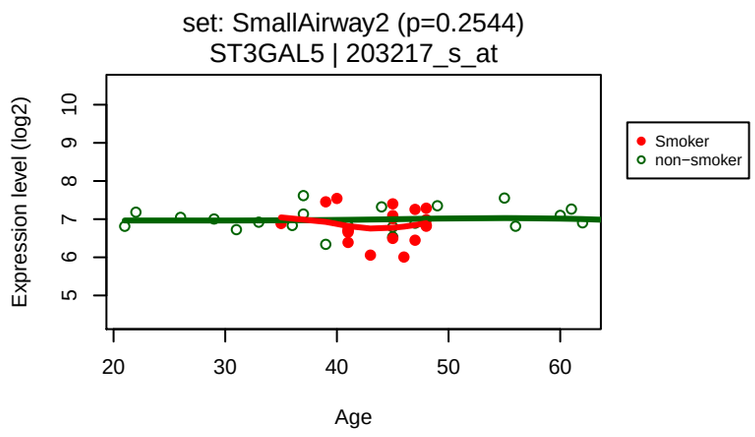
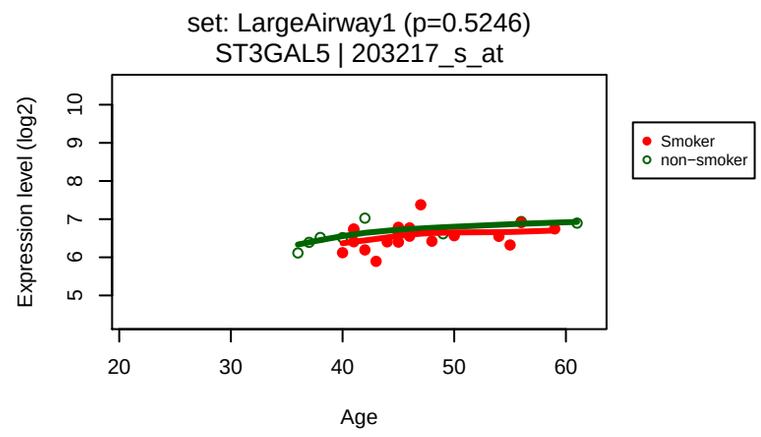

Gene expression of smokers and nonsmokers
gene: ST3GAL5 | 203217_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4341
p=0.2544
p=0.0239
p=0.5246
p=0.4824
p=0.8361
p=0.7606
p=0.2401
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

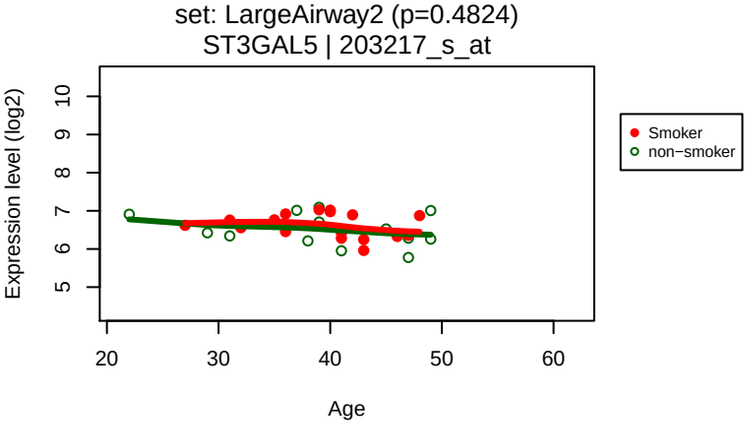

Gene expression ratio (smokers/nonsmokers)
gene: ST3GAL5 | 203217_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4341
p=0.2544
p=0.0239
p=0.5246
p=0.4824
p=0.8361
p=0.7606
p=0.2401
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

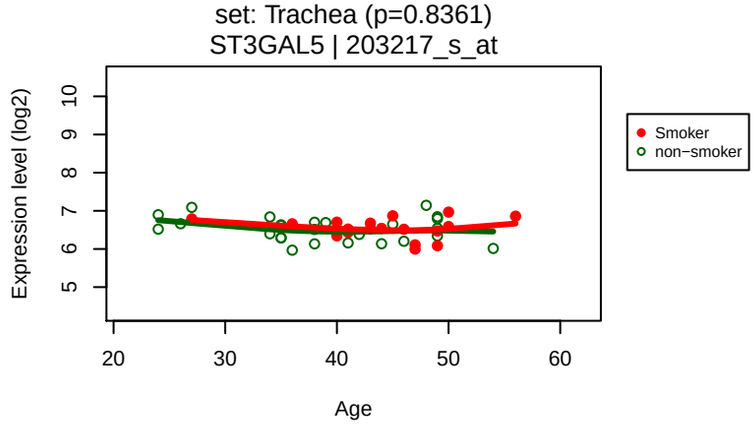

Gene expression ratio in Males
gene: ST3GAL5 | 203217_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.499
p=0.8579
p=0.6233
p=0.3363
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ST3GAL5 | 203217_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3558
p=0.2061
p=0.8692
p=0.6166
SmallAirway2
LargeAirway2
Trachea
Macrophage2

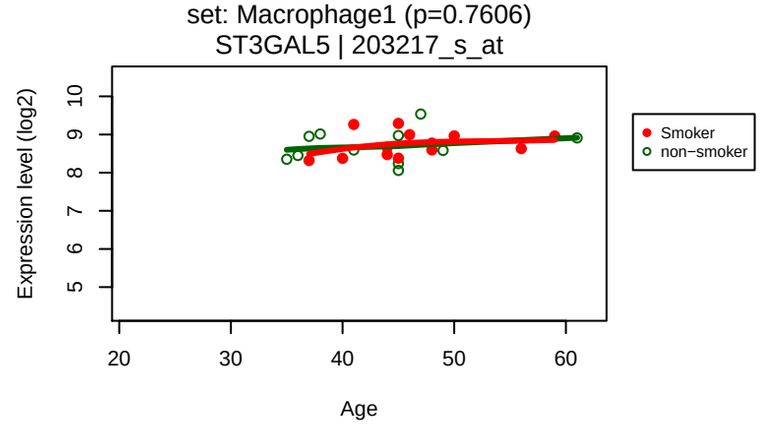

set: SmallAirway1 (p=0.4341)
ST3GAL5 | 203217_s_at
Expression level (log2)
Age
Smoker
non−smoker

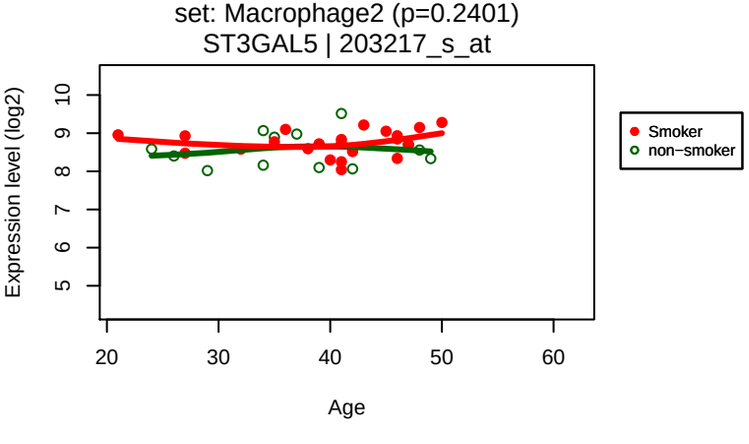

set: SmallAirway2 (p=0.2544)
ST3GAL5 | 203217_s_at
Expression level (log2)
Age
Smoker
non−smoker

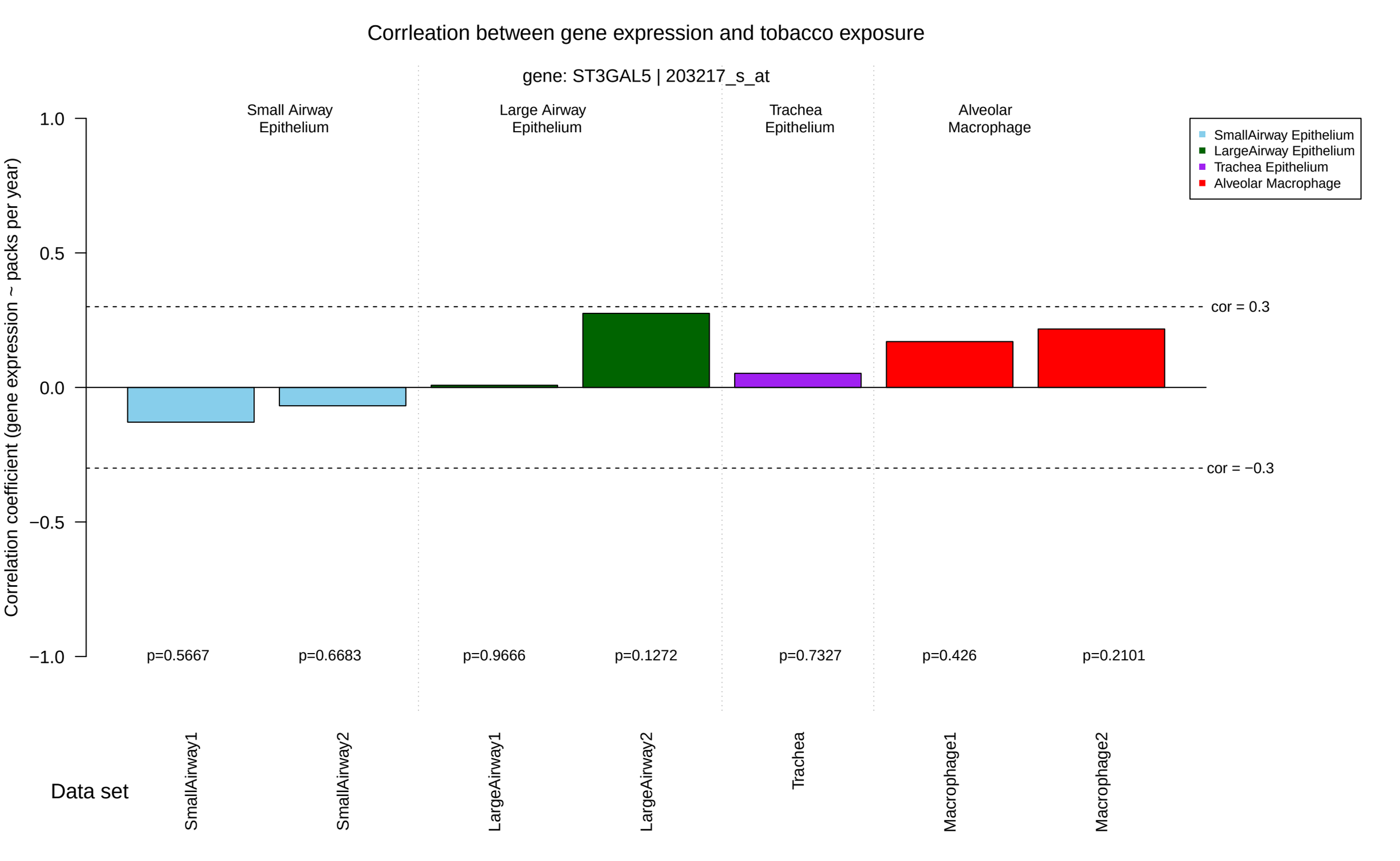

set: LargeAirway1 (p=0.5246)
ST3GAL5 | 203217_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4824)
ST3GAL5 | 203217_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8361)
ST3GAL5 | 203217_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7606)
ST3GAL5 | 203217_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2401)
ST3GAL5 | 203217_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ST3GAL5 | 203217_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.5667
p=0.6683
p=0.9666
p=0.1272
p=0.7327
p=0.426
p=0.2101
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2